

## **Topic: Acute Respiratory Distress for Syndrome in Septic ICU Patients: Demographic Risk, Severity, and Clinical Outcomes in MIMIC-IV**

### **Introduction**

Acute Respiratory Distress Syndrome (ARDS) is a severe form of acute hypoxemic respiratory failure characterized by diffuse lung injury and bilateral infiltrates unrelated to cardiac causes. It remains a major source of morbidity and mortality in the ICU, particularly among patients with sepsis, one of its most common precipitating factors.<sup>1</sup> Early recognition and risk stratification are essential because ARDS often requires prolonged mechanical ventilation and carries high in-hospital mortality.

The Berlin criteria classify ARDS severity using the  $\text{PaO}_2/\text{FiO}_2$  (PF) ratio: mild (200–300), moderate (100–200), and severe ( $<100$ ), with mortality rising across categories.<sup>2</sup> Understanding how severity relates to clinical outcomes in real-world ICU settings can support decision-making and resource allocation. Several clinical and behavioral risk factors increase susceptibility to ARDS, including older age, tobacco exposure, and alcohol use.<sup>(3-5)</sup> Emerging literature also suggests that demographic and social determinants, such as race and insurance status, may influence ARDS development and outcomes.<sup>6</sup> Characterizing these factors in large ICU datasets may help identify vulnerable subgroups.

Beyond baseline characteristics, early PF ratio trajectories in the first 48 hours after ARDS onset: improving, stable, or worsening, have been shown to predict mortality and ventilation duration.<sup>7</sup> Examining these trajectories may offer additional insight into patient prognosis.

Using MIMIC-IV, this study examines ARDS among septic ICU patients with three objectives:

1. Identify demographic, social, and behavioral factors associated with ARDS development.
2. Assess the association between ARDS severity and in-hospital mortality, including differences across ventilation status and ICU types.
3. Evaluate early PF ratio trajectories and their relationship with in-hospital mortality and ventilation duration.

## Method

### Demographic Analysis

First, demographic information for all patients was extracted from the MIMIC-IV icustays, admissions, patients, and diagnoses\_icd tables using a series of SQL queries. The demographic variables of interest included insurance status, language, marital status, race, anchor age, and gender. Along with these variables, standard demographic information of the patients such as subject\_id, hadm\_id, and stay\_id was collected. Since this study focuses on the occurrence of ARDS per stay, stay\_id was treated as the primary key. In addition to demographic data, the analysis also considered the patient's lifestyle values listed in the table below. These values were extracted from the chartevents table through SQL queries and then further joined to the demographic dataset on stay\_id using a Python pandas DataFrame.

Table 1. Variables, entries and applicable codes

| Description           | Entries                                      | Code   |
|-----------------------|----------------------------------------------|--------|
| Tobacco use           | stopped, current, former, never              | 227687 |
| Alcohol use           | 0/1                                          | 225106 |
| Recreational Drug Use | 0/1                                          | 225110 |
| living_status         | Lives alone/ Other/ With family/ With spouse | 225072 |
| assistive_devices     | Cane/None/Other/Wheelchair/Walker            | 225101 |
| self_adl              | 0/1                                          | 225092 |
| appetite              | Deferred/Fair/Good/Not assessed/ Poor        | 225120 |

Using stay\_id as the primary key, we retrieved PaO<sub>2</sub> (Item ID 220224) and FiO<sub>2</sub> (Item ID 223835) values.

To provide a holistic understanding of the values, the minimum and maximum values for both parameters during each stay were calculated and appended to the final dataset. Given that ARDS is defined by the indicator function  $ARDS = 1(\frac{PaO_2}{FiO_2} \leq 300)$ , we restricted the analysis to paired readings recorded within one minute of each other to ensure clinical relevance. As anticipated, a significant number of ICU stays are missing these measurements. To reduce selection bias, assuming that only patients with ARDS

symptoms typically had these types of measurements taken, we labeled missing entries as negative (i.e., assumed no ARDS). While this approach carries a slight risk of misclassification, we assumed that  $\text{PaO}_2$  and  $\text{FiO}_2$  monitoring is standard protocol for any patient suspected of ARDS. Therefore, we assumed that all patients who genuinely suffered from ARDS during the duration of their stay had these measurements recorded.

To train a predictive classifier model, we strip deterministic columns (such as min  $\text{PaO}_2$  and max  $\text{FiO}_2$ ) from the table and split the table 80:20 for training and testing. The models are trained using the training set with the ARDS labels as ground truth. We tested several models (Logistic regression, XGBoost, CatBoost, LightGBM, MLP, and Random Forest) to find that Random Forest is the best predictive model for our case. We used sklearn's Random Forest Package with the following as parameters:

`n_estimators=500, max_depth=None, random_state=42, and class_weight='balanced'.`

### **ARDS Severity Associations**

Using the same cohort as the demographic analysis section, the lowest patient  $\text{PaO}_2 / \text{FiO}_2$  (pf) ratios were calculated per `hadm_id` and were then placed on a box plot with in-hospital death. To get an idea of how well the Berlin criteria reflects in-hospital death, patients were categorized into different ARDS severity strata: no ARDS ( $>300$ ), mild ( $200-300$ ), moderate ( $100-200$ ), and severe ( $<100$ ). A table was generated showing the ARDS severity, number of hospital admissions, and in-hospital death percentage.

To examine trends of ARDS severity, ventilation status, and in-hospital death percentage, a table was made in the following manner. Similar to before, patients were categorized into different ARDS severity strata based on `hadm_id`. We then grouped patients by different ventilation status, simplifying to “Yes”, “No”, “Mixed”, and “NA”. Ventilation status was taken at the timepoint immediately after the lowest pf ratio was recorded, as that would imply intervention was taken at the most extreme course of the patient's stay. Additionally, ventilation status may have different values if there were multiple `stay_ids` per `hadm_id`, so the “Yes” category means that patients always received ventilation support, “No” category

means that patients never received ventilation support, “NA” category means that ventilation status was not recorded, and “Mixed” category means that patients had any combination of the three previous categories. Once patients were grouped by ARDS severity and ventilation status, in-hospital death percentage and number of hospital admissions were calculated for each group.

Lastly, we wanted to see whether the ICU type had a significant impact on in-hospital mortality. To visualize this, a bar plot was generated. In this bar plot, the y-axis represents the different ICU types and the x-axis represents the number of admissions based on `hadm_id`. Within the different ICU types, when patients had multiple/mixed ICU stays, they were categorized into a new group called “multiple/mixed ICU stays”. Each bar shows the survival/death ratio (blue and red, respectively) with the percent survival appended to the right of the bar.

## **Trajectory Analysis**

We characterized the evolution of respiratory function during the first forty-eight hours after ARDS onset by constructing individual  $\text{PaO}_2$  to  $\text{FiO}_2$  trajectories and applying a combined rule-based and clustering-based approach to classify trajectory patterns. ARDS onset for each ICU stay was defined as the earliest time at which the  $\text{PaO}_2$ -to- $\text{FiO}_2$  ratio was 300 or lower while the patient was receiving mechanical ventilation. All  $\text{PaO}_2$  to  $\text{FiO}_2$  measurements recorded during ventilation from 0 to 48 hours after this onset time were retained for trajectory construction.

Each measurement was assigned to the nearest four-hour interval relative to the onset to create a standardized temporal structure with twelve aligned time points. When multiple measurements mapped to the same interval, we selected the value with the smallest deviation from the target time. Stays were included only if at least four time points had observed values. Within-row interpolation was applied to fill missing PF values across the twelve intervals, producing a complete trajectory for each stay.

Trajectory classification proceeded in two stages. First, we computed the difference between the PF ratio at onset and that at 48 hours. Stays with a non-positive difference were categorized as worsening to

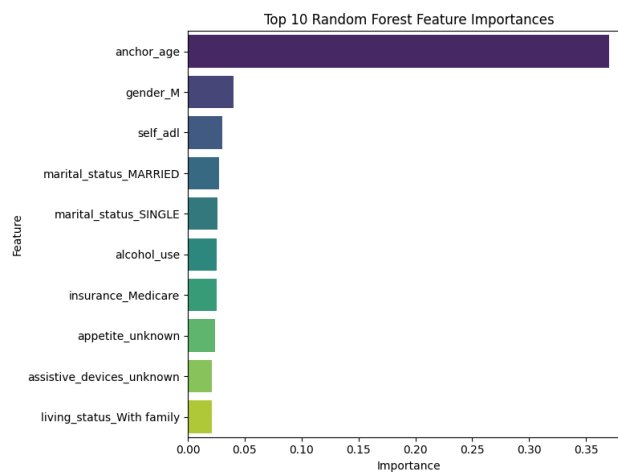
represent patients whose oxygenation failed to improve. Stays with a positive difference were eligible for clustering. In these cases, the 12 PF values were standardized using z-score normalization, and a k-means clustering with 2 clusters was applied. The two resulting clusters were ordered by the mean change in PF ratio over the 48-hour window. The cluster with the smaller mean improvement was labeled Improving slow, and the cluster with the larger mean improvement was labeled Improving fast.

After assigning each stay to one of the three groups, we computed summary measures, including in-hospital mortality, median ventilation duration, and median minimum PF ratio, to compare outcomes across trajectory patterns. Group-level trajectories were visualized by plotting the average PF ratio at each aligned time point, and individual trajectories were displayed for a random subset of stays to illustrate within-group variability.

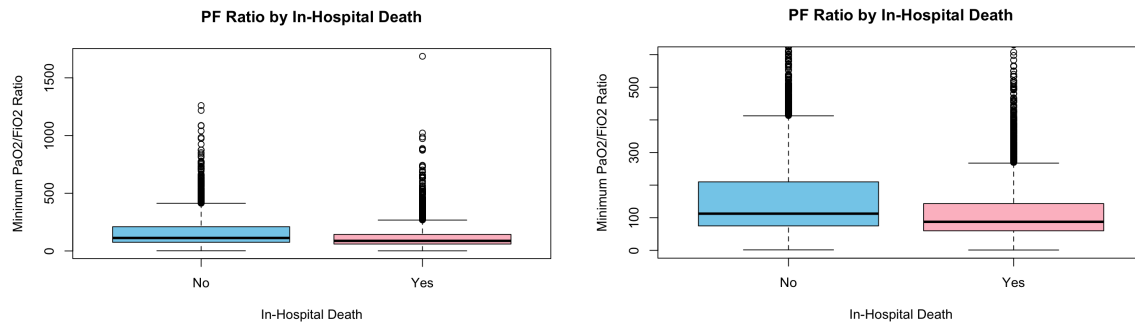
## Results

### Demographic Analysis

The Random Forest model achieved an accuracy of 0.7351 and an AUROC of 0.7889, which is a decent model that will be able to help predict the occurrence of ARDS simply through the patient's background information. Additionally, the ranking of feature importance is as follows:



## ARDS Severity Association



In the above plots (zoomed in plot on the right) showing minimum pf ratio and in-hospital death, we can see that patients admitted with sepsis generally fall within the values for the Berlin criteria to be applicable, with visible outliers only being a small percentage of the cohort size. We can see that cases where in-hospital death occurs generally have a lower minimum pf\_ratio.

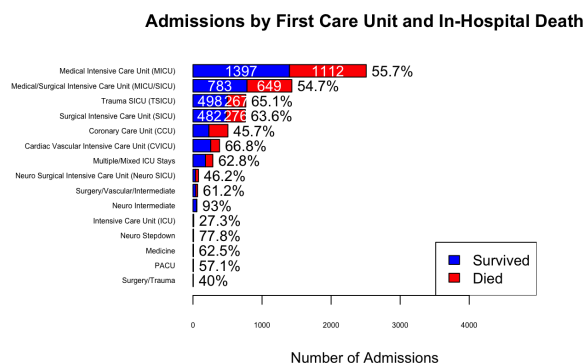
| ards_severity<br><chr> | total_admissions<br><chr> | inhospital_death_pct<br><chr> |
|------------------------|---------------------------|-------------------------------|
| Severe                 | 3612 (52.4%)              | 49.7%                         |
| Moderate               | 1923 (27.9%)              | 38.1%                         |
| Mild                   | 698 (10.1%)               | 30.9%                         |
| No ARDS                | 654 (9.5%)                | 25.1%                         |

In the above table, we can very clearly see a significant difference in in-hospital death percentage across ARDS severities, with in-hospital death percentage increasing with ARDS severity.

| ards_severity<br><fctr> | ventilation_status<br><fctr> | inhospital_death_pct<br><chr> | total_admissions<br><chr> | ards_severity<br><fctr> | ventilation_status<br><fctr> | inhospital_death_pct<br><chr> | total_admissions<br><chr> |
|-------------------------|------------------------------|-------------------------------|---------------------------|-------------------------|------------------------------|-------------------------------|---------------------------|
| Severe                  | Yes                          | 31.7%                         | 2256 (32.8%)              | Mild                    | Yes                          | 14.1%                         | 461 (6.7%)                |
| Severe                  | No                           | 0.0%                          | 2 (0.0%)                  | Mild                    | No                           | NA%                           | 0 (0.0%)                  |
| Severe                  | Mixed                        | 56.4%                         | 165 (2.4%)                | Mild                    | Mixed                        | 26.7%                         | 15 (0.2%)                 |
| Severe                  | NA                           | 82.9%                         | 1189 (17.3%)              | Mild                    | NA                           | 66.2%                         | 222 (3.2%)                |
| Moderate                | Yes                          | 21.5%                         | 1268 (18.4%)              | No ARDS                 | Yes                          | 12.6%                         | 429 (6.2%)                |
| Moderate                | No                           | NA%                           | 0 (0.0%)                  | No ARDS                 | No                           | NA%                           | 0 (0.0%)                  |
| Moderate                | Mixed                        | 50.0%                         | 52 (0.8%)                 | No ARDS                 | Mixed                        | 25.0%                         | 8 (0.1%)                  |
| Moderate                | NA                           | 71.8%                         | 603 (8.8%)                | No ARDS                 | NA                           | 49.8%                         | 217 (3.2%)                |

In the above table, we can see that there are a lot of admissions that fall under the “NA” category. As for what the “NA” category means here, we believe that it means that the patient did not receive any ventilation support, as the “No” category only has two occurrences. Regardless, we can see that

ventilation support significantly improves patient outcome, as across all ARDS severities, the groups that had the lowest in-hospital death percentage were labelled “Yes” under ventilation\_status.



Lastly, the above barplot shows ICU types, number of admissions, and patient survival rate. Overall, it seems as though the ICU type the patient is admitted to has little effect on survival rate.

## Trajectory Analysis

Figure 1 displays three  $\text{PaO}_2/\text{FiO}_2$  trajectory groups. The Worsening group shows a decline in mean PF ratio from approximately 185 to 130, consistent with persistent respiratory failure. The Improving slow group shows a gradual increase from about 90 to 155, whereas the Improving fast group increases rapidly from about 135 to more than 225.

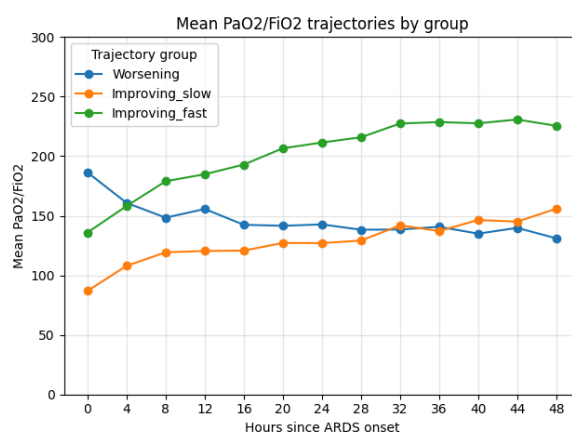


Figure1: Mean  $\text{PaO}_2/\text{FiO}_2$  trajectories during the first forty eight hours after ARDS onset

Figure 2 shows individual and average  $\text{PaO}_2/\text{FiO}_2$  trajectories for each group. In each panel, the light thin lines represent twenty randomly selected patients, and the bold line represents the group mean. In the

Worsening group, most individual trajectories remain low or decline, whereas in the Improving slow and Improving fast groups they show gradual and rapid increases, respectively. The mean trajectories are consistent with the patterns in Figure 1 and illustrate the degree of within group variability around each average curve.

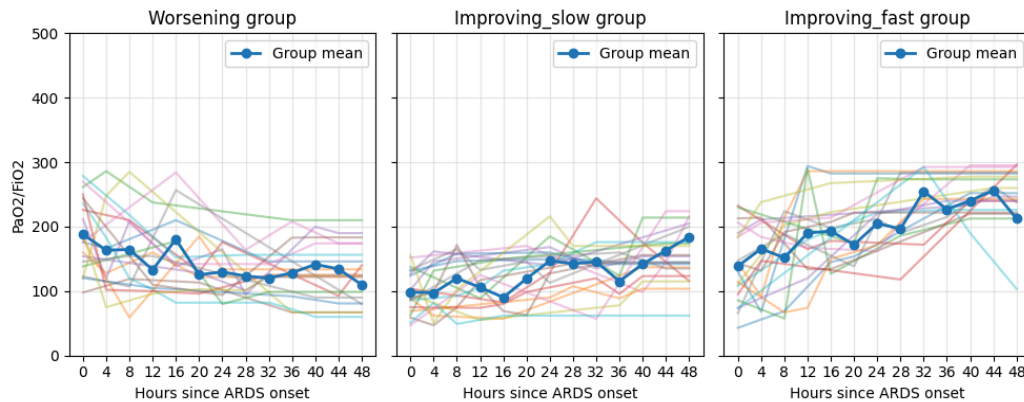


Figure2: Individual variability and mean  $\text{PaO}_2/\text{FiO}_2$  trajectories within each trajectory group

Figure 3 compares clinical outcomes across the three trajectory groups. Mortality differed across patterns of respiratory recovery, with the Worsening group showing the highest in hospital mortality, followed by the Improving slow group, and the Improving fast group showing the lowest mortality. Median ventilation duration showed less separation across groups, although the Worsening group had the shortest duration and both improving groups had slightly longer median ventilation times. These patterns align with the trajectory based differences observed earlier and suggest that early improvements in oxygenation are associated with more favorable survival outcomes.

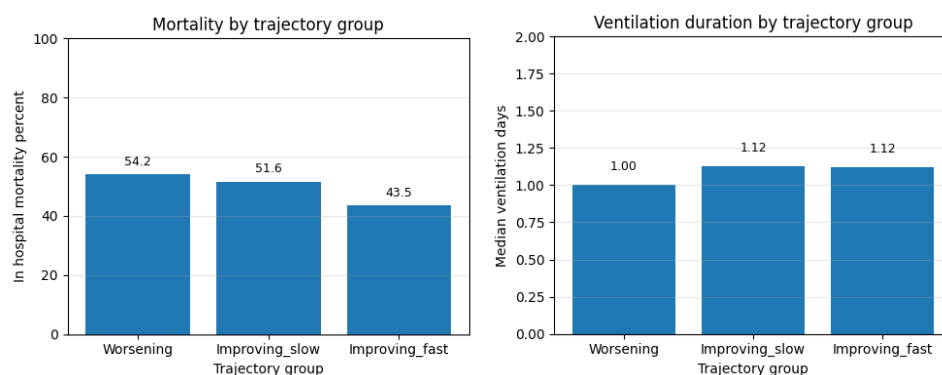


Figure3: Mortality and ventilation duration across  $\text{PaO}_2/\text{FiO}_2$  trajectory groups



## **Discussion**

Age emerged as one of the strongest predictors of ARDS, consistent with prior literature showing that older adults have higher susceptibility and mortality due to decreased physiologic reserve and greater comorbidity burden.<sup>4</sup> Because many age-related conditions also increase ARDS risk, isolating age effects remains challenging. Males were more likely to develop ARDS, though mortality differences by gender were minimal, which aligns with studies showing higher incidence but similar outcomes once ARDS develops.<sup>5</sup> Medicare insurance appeared as an important feature, likely reflecting its role as coverage for older adults or those with disabilities, rather than an independent socioeconomic effect.

We observed a clear gradient in mortality across ARDS severity: 49.7% in severe ARDS, 38.1% in moderate, and 30.9% in mild, reinforcing the prognostic value of Berlin severity categories. Mortality in septic patients without ARDS (25.1%) remained high, but the incremental rise with increasing ARDS severity underscores the added burden of lung injury beyond sepsis alone. No meaningful differences in in-hospital mortality were seen across first-care ICU types. This may reflect early transfers, consistent ICU-level adherence to ARDS management guidelines, or the predominance of patient-level severity over unit-specific factors.

Early  $\text{PaO}_2\text{:FiO}_2$  (PF) ratio trajectories carried important prognostic information. Patients with worsening oxygenation had the highest mortality (54.2%), whereas rapidly improving trajectories showed lower mortality (43.5%). Although differences in ventilation duration were modest, these findings support evidence that dynamic changes in oxygenation could be more informative than single-time-point PF ratio measurements.

## **Conclusion**

These results reinforce established ARDS risk factors, highlight the strong association between severity and mortality, and suggest that early PF trajectories may aid in identifying high-risk patients. Limitations

include use of a single-center dataset, potential misclassification of ARDS based on physiologic criteria, and the possibility that insurance type acts as a proxy for age or comorbidity.

## **Contribution (describe individual contributions to the project)**

- **Nat Chairuengjitjaras:** Formulated project questions; selected analytic variables; conducted data quality checks; interpreted results; and led the writing of the Introduction, Discussion, and Conclusion sections.
- **Andrew Yu:** Performed the stratified analyses examining ARDS severity and in-hospital mortality across ICU types and demographic subgroups.
- **Mengyao Wang:** Conducted the PF ratio trajectory analysis, including trajectory grouping and outcome comparisons.
- **Vincent Angelo:** Completed the demographic and baseline characteristics analysis and contributed to descriptive data interpretation.

## **References**

1. **Bellani G, Laffey JG, Pham T, et al.** Epidemiology, patterns of care, and mortality for patients with acute respiratory distress syndrome in intensive care units in 50 countries. *JAMA*. 2016;315(8):788-800. doi:10.1001/jama.2016.0291
2. **Ranieri VM, Rubenfeld GD, Thompson BT, et al.** Acute respiratory distress syndrome: the Berlin definition. *JAMA*. 2012;307(23):2526-2533. doi:10.1001/jama.2012.5669
3. **Zhang L, Xu J, Li Y, Meng F, Wang W.** Smoking on the risk of acute respiratory distress syndrome: a systematic review and meta-analysis. *Crit Care*. 2024;28(1):122. Published 2024 Apr 14. doi:10.1186/s13054-024-04902-6
4. **Brown R, McKelvey MC, Ryan S, et al.** The Impact of Aging in Acute Respiratory Distress Syndrome: A Clinical and Mechanistic Overview. *Front Med (Lausanne)*. 2020;7:589553. Published 2020 Oct 26. doi:10.3389/fmed.2020.589553
5. **McNicholas BA, Madotto F, Pham T, et al.** Demographics, management and outcome of females and males with acute respiratory distress syndrome in the LUNG SAFE prospective cohort study. *Eur Respir J*. 2019;54(4):1900609. Published 2019 Oct 17. doi:10.1183/13993003.00609-2019
6. **Erickson SE, Shlipak MG, Martin GS, et al.** Racial and ethnic disparities in acute lung injury epidemiology and outcomes: a multicenter study. *Chest*. 2009;136(5):1304-1311. doi:10.1378/chest.09-0798
7. **Li M, Liu F, Yang Y, et al.** Identifying vital sign trajectories to predict 28-day mortality of critically ill elderly patients with acute respiratory distress syndrome. *Respir Res*. 2024;25(1):8. Published 2024 Jan 4. doi:10.1186/s12931-023-02643-8